MECHANO-METABOLISM FINITE-ELEMENT PREDICTIONS IN LUMBAR DISC DEGENERATION AND LOW BACK PAIN STRATIFICATION

Estefano Muñoz-Moya[1], Terence McSweeney[2], Jaro Karppinen[2], Carlos Ruiz Wills[1], Mark van Gils[3], Gemma Piella[1], Jerôme Noailly[1]

[1] BCN MedTech, Universitat Pompeu Fabra, Barcelona, Spain, estefano.munoz@upf.edu.
[2] Research Unit of Health Sciences and Technology, University of Oulu, Oulu, Finland
[3] Faculty of Medicine and Health Technology, Tampere University, Tampere, Finland

Introduction

Low back pain intensity (LBPi) and intervertebral disc (IVD) degeneration (IDD) are multifactorial conditions where imaging, mechanics, and bio–psycho–social factors interact [1]. Lactate has been proposed as an early in-vivo IDD marker [2], with intradiscal transport and morphology suspected to control the metabolic stress on IVD cells [3]. Our goal is to test if the metabolic stress constitutes a relevant risk factor for stratifying LBPi and IDD in the Northern Finland Birth Cohort (NFBC) [1]. We used a metabolic stress finite-element (FE)-derived risk axis (x_{risk}), obtained from patient-specific (PP) mechano-metabolic simulations [5], and transferred the risk to the Disease State Fingerprint (DSF) model [4] as a mechanical biomarker.

Methods

The SpineView cohort [3] (N=169 PP IVD models, under daily mechanical loading over 3 simulated days) was the mechanistic reference. Fully coupled oxygen–glucose–lactate transport equations were solved to predict nucleus pulposus (NP) cell viability [5]. At day 3, 32 morphophysics predictors were built (20 morphometrics [3] + 12 solute/pH readouts across posterior, central, and anterior transition zone, ATZ), and a continuous risk score (S_{risk}, higher = worse) was learned by logistic regression and stratified by Fisher–Jenks clustering (Low/Mid/High). Performance was assessed by repeated 5-fold cross-validation; feature–risk relations were profiled with SHAP and Spearman analyses. A Support Vector Regression (SVR) surrogate was then trained to predict the risk score from IVD morphology (train/validation/test 60/20/20) and applied to N=516 reconstructed L4–L5 discs from NFBC MRI (Pixel2Mechanics/mesh morphing [3]), yielding S_{risk}^{NFBC}. This score was incorporated as a new Mechanistic Risk into the DSF for LBPi and IDD (Pfirrmann>3) in NFBC.

Results

The mechano-metabolic axis separated the 13/169 discs with simulated NP cell death (Figure 1) with strong discrimination (full-fit RAUC 0.997; CV RAUC 0.95±0.08), and Fisher–Jenks concentrated all positive discs in the High stratum (Figure 1). ATZ solute/pH features dominated the separation, with ATZ lactate rising near-linearly from Low to Mid to High (Spearman ρ=+0.93, r^2=0.89). Anterior AF sagittal area and disc mid-height — most altering anterior transport paths under load — ranked among the top SHAP drivers of ATZ solutes, linking morphology to waste accumulation. The SVR surrogate reproduced the FE-derived risk score (r^2=0.97) via the same morphological features (annulus volume, disc mid-height, anterior sagittal area). DSF showed that LBPi is driven mainly by psychosocial and stress-related variables, whereas level-specific IDD is dominated by MRI-derived radiomics and structural flags. The Mechanistic Risk showed non-negligible relevance in the L4–L5 IDD and LBPi DSF, on the morphological/radiomic branch otherwise treated as purely data-driven.

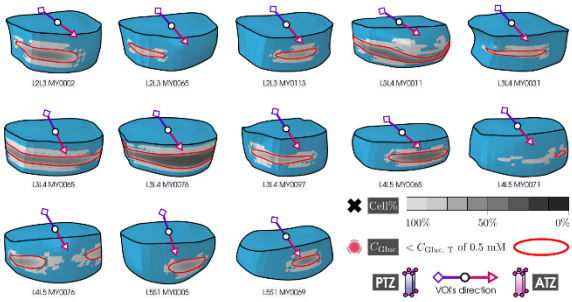


Figure 1: 13/169 discs with NP cell death.

Conclusions

A single supervised risk axis yields a clinically interpretable readout: higher mechano-metabolic risk points to higher ATZ lactate, complementing MR spectroscopy evidence that lactate precedes matrix changes in IDD[3]. Because x_{risk} shares its morphological descriptors with the radiomics and MRI structural flags dominating NFBC, its transfer shows that FE-based mechanistic modelling can help explain those imaging features: morphology shapes anterior solute-transport paths and, thereby, metabolic waste accumulation and cell viability, which imaging captures indirectly. This framework connects personalized chemo-mechanical modelling with interpretable, population-level bio–psycho–social stratification. Validation with external cohorts and lactate-sensitive MRS acquisitions is ongoing, toward clinic-ready biomarkers.

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